A3 IN-GRD CTL & INTERPOLATOR

File: 3458A A3 66533 2021 KS SMD p1.kicad_sch

A3 A/D CIRCUIT

File: 3458A A3 66533 2021 KS SMD p2.kicad_sch

Sheet: /
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Title: 3458A A3 A/D CIRCUIT 03458-66533

Size: A5	Date: 2026-04-11	**Rev: 002**
KiCad E.D.A. 9.0.7		Id: 0/4

CONNECTOR TO AC & PS BOARD

P1: ACSTB_F, ACDO_F, ACCK_F, MSRST, +5, XT/H 2, −18, +18, −21, LEVEL 2, XPOVLD, PS_SENSE, XNOVLD, AC_AD 3, ACDR_F

CONNECTOR TO DC BOARD

P2: +5_F, +18, −18, XO_OVLD, DCSTB_F, DCDA_F, DCCK_F, XMC_F, MZ_F, PC_F, MSRST_O, DCDR_F, XSYNC_F

MH201 MOUNTING, MH202 MOUNTING, MH203 MOUNTING, MH204 MOUNTING, MH205 MOUNTING

R281 1.0k XMC, R282 1.0k MZ, R283 1.0k PC, R270 1.0k DCSTB, R271 1.0k DCCK, R272 1.0k DCDA, C272 330p

R273 1.0k DCDR, C273 100p

TP17 +5, L220 100n, +5_a, TP20 GND

+5_a, JM210, +5_b

TP18 +18, C913 100n, +18, C252 4.7µ, −18, C253 4.7µ, TP19 −18

R264 1.0k ACDO, R260 1.0k ACSTB, R261 1.0k ACCK, R262 1.0k ACDA, C262 330p

R280 1.0k XO_OVLD, C280 10n

R263 1.0k ACDR, C263 100p

C914 100n, C250 4.7µ, L250 10µ, +5_F, C251 4.7µ

R200 10.0k +5, R201 51.1k XF/R, XF/R_F, C200 100n

MEASUREMENT SYNC, TP210, R285 10.0k XSYNC_F, XSYNC

MSRST_O, R284 10.0k, MSRST_F

MSRST, R286 10.0k, C284 10n

CR241 MMBD1501, FOVLD

R240 51.1k +5, R241 10.0k XPOVLD, R242 150, POVLD, C240 100p, CR240 1N5711, XENOV

R245 51.1k +5, XNOVLD, R246 10.0k, NOVLD, C245 100p, CR245 1N5711, XENOV

SIGNAL FILTERING

EXT. IN/OUT FIBER OPTICS

TP13 EXT_OUT, U353A, U353B, U353C, U353D, U353E, 74HCT14D

EXT_OUT, R354 150, CR352 1N4150, U354 GRAY, HFBR1521Z

U351, HFBR2521Z BLUE, +5_b, R351 1.5k, C351 100n, U353F, TP14 EXT_IN, EXT_IN

U353G, GND VCC 14, C353 33µ, C904 100n, 74HCT14D

CROSS-GUARD FIBER OPTICS

TP15 XGRD_OUT, XGRD_OUT, U302A, CR301 1N5711, R302 499, R303 750, U302B, C302 22p, 74HCT14D

U303A, U303B, U303C, U303D, U303E, U303F, 74HCT14D

+5_d, L303 10µ +5, C303 33µ, R304 150, CR302 1N4150, U304 GRAY, HFBR15212

+5_d, C301 100n, R301 1.5k, U301, HFBR2521Z BLUE, U302D, TP16 XGRD_IN, XGRD_IN

U302G, GND VCC 14, C908 100n, 74HCT14D

U303G, GND VCC 14, C907 100n, 74HCT14D

TIME INTERPOLATOR

CR405 ZPD3.0, CR406 ZPD3.0, +18, −18, +15, −15

+15, CR4051 ZPD3.0, C4050 100n, 78L05, U4051, C4055 2.2µ, C4056 100n, +5_T

C4059 100n, U4052, C4058 2.2µ, C4057 100n, CR4052 ZPD3.0, 79L05, −5_T

66523: w/o 0R

HD1ONS, HD2ONS, HD4ONS, HD8ONS

R401-A1 453k, R402-A1 226k, R403-A1 113k, R404-A1 56.2k

R401-B1, R402-B1, R403-B1, R404-B1

R405-A1 18.7k, R405-B1, R406-A1 14.0k, R406-B1

+5_b, C404 100p, +5_T, U405 TL3016, TP406 HTRIG, HTRIG 2, C403 22p, C414 100p, −5_T

+5_b, C909 100n, 74HC4094D, ACSTB, ACCK, ACDI, SER IN, ENMZ 2, FOVLD, ACDO

R412 56.2k, R411 113k, R410 226k, R409 453k, R408 909k

+18, C400 100n, R413 5.62k

+18, C413 100n, U402 OP07C, R415 100, Q401 MMBT3906, Q402 MMBT3906, R416 649

+18 +5_b, R417 100, C405 100n, C910 100n

+15, C407 100n, U403E DG201HS, C406 100n, −15

R407 47

HRAMP, CR401 BAR18, TP404 5mA, CR402 1N4150

TP402 LRESET, TP403 HSTORE, TP401 LRAMP, 74HC368D, U400A, U400B

+5_b, C415 100n, U400C 74HC368D

U403C DG201HS, U403D DG201HS, C408 1n, CR403 1N4150, U403B DG201HS, R418 10

Q403 PF5301, C409-A1 470p, C409-B1 10p, Σ480p, 66523: 1x510p, C410 100n

+15, C411 100n, U404 LT1122, TP405 RAMP, C412 100n, −15

IN GUARD CONTROL LOGIC

+5_c, +5, L230 100n, U230, VDD, GND, OUT, 20MHz, C130 100n

L233 100n, CK10S, U131B 74F112D, L221 100n, CK10

L232 100n, +5_b, XCK20 3

L231 100n, XCK20_CPLD 2

J2, X1, XCK20_OUT 2

+3V3, L223 27µ, C902 100n, VDD

R220-1 33k, R220-2, R220-3, R220-4, R220-5, R220-6, R220-7, R220-8 33k

CPLD & LEVEL SHIFT (Seite 2)

U220 AT89LS51: P3.7/RD, P0.0/AD0, P0.1/AD1, P0.2/AD2, P0.3/AD3, P0.4/AD4, P0.5/AD5, P0.6/AD6, P0.7/AD7, P2.3/A11, P3.0/RXD, P3.1/TXD, P3.2/INT0, P2.7/A15, P2.6/A14, P2.5/A13, P2.4/A12, P1.1, P1.2, P1.3, P1.4, P1.5, P1.6, P1.7, P3.4/T0, P3.5/T1, RST, XTAL2, XTAL1

_CMDI, _ADD0, _ADD1, _ADD2, _ADD3, _XRXW, _NRDGOUT, _BFSTAT, _OTHINT, _IOSTB, _UPDA, _IOCLK, _XINTR, _XETRD, _HOLD, _SEND, _XEXOR, _ADBSY, _INCMP, _XCINCMP, _UPTRG, _ENTRG, _XCOT0, _XCOT1, _XNRFT1, _COMIN, _C1MIN, _RST 2, _CK10_uC, MOSI, MISO, SCK

_XO_OVLD_F, _XMC, _MZ, _PC, _LEVEL, P3.6/WR, EA/VPP, ALE/PROG, PSEN

R223 1.0k, TP220 ADBSY

+5_b, U131C 74F112D, C131 100n, C906 100n

+3V3, U213E 74LCX00, C905 100n

L222 27µ, R4 NetTie

TP11 CK10, 74LCX00, U213A, U213D, U213C, _CK10_uC, _CK10_CPLD 2

JD2, +3V3, MISO, SCK, _RST, MOSI, 3.3V ISP

UNUSED PARTS

U131A 74F112D, +5_b

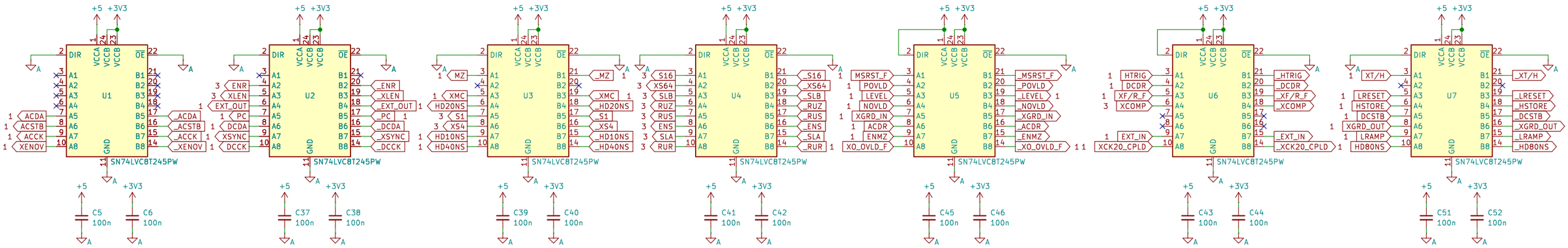
74LCX00, U213B

U403A DG201HS

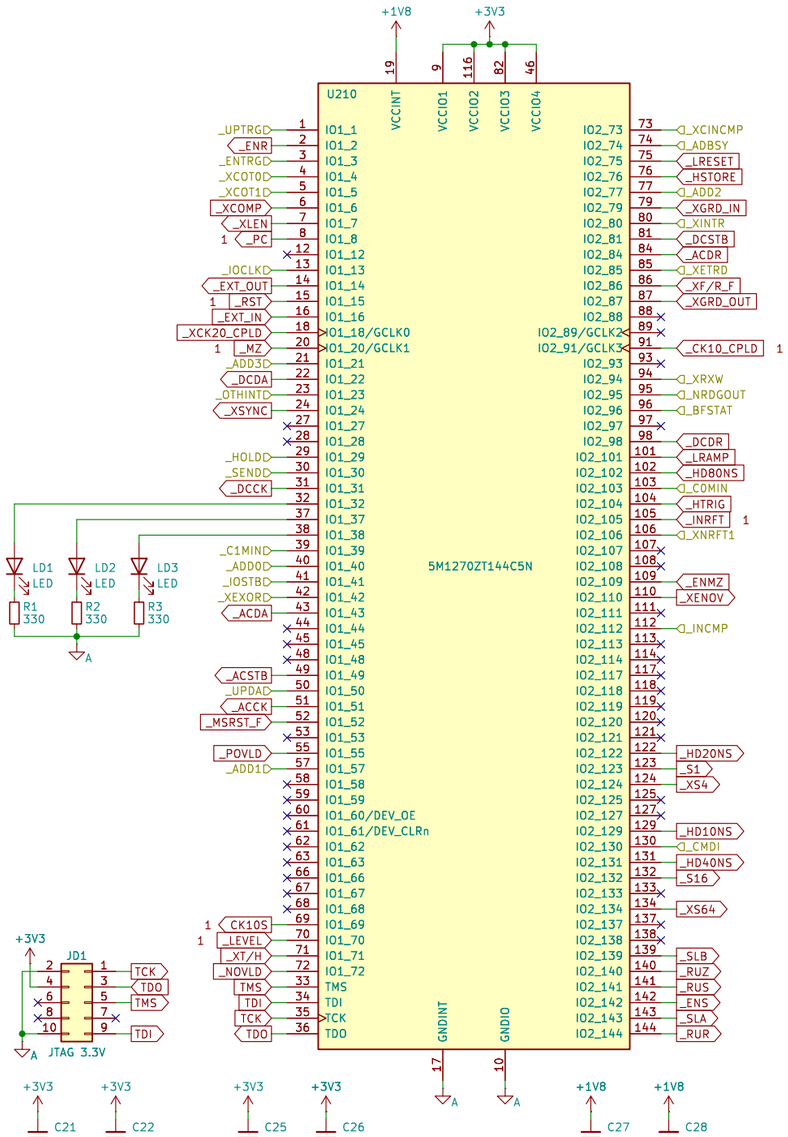
U302F, U302C, U302E, 74HCT14D

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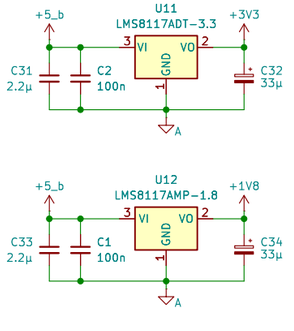
Level shifter



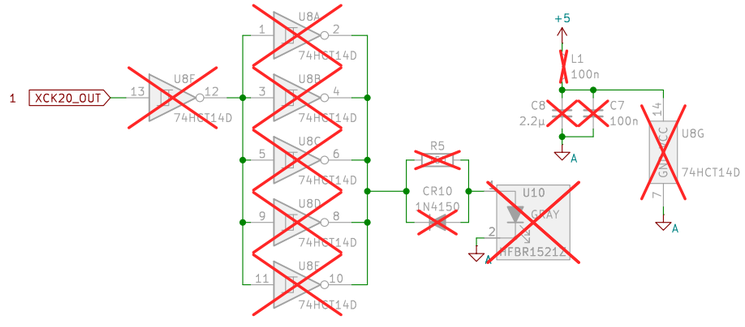
CPLD



Power supply CPLD, µC & Level shifter



EXT CLOCK SYNC OPTION



Sheet: /A3 IN-GRD CTL & INTERPOLATOR/CPLD & LEVEL SHIFT (Seite 2)/	
File: 3458A A3 66533 2021 KS SMD p1.1.kicad_sch	
Title: 3458A A3 CPLD & LEVEL SHIFT 03458-66533	
Size: A3	Date: 2026-04-11 — **Rev: 002**
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Part#	Rev	ERC	Date (highest DC)	Changes to previous revision (maybe incomplete)
66503	A	2912	14–1989 (highest DC)	First Revision
66503	B	2931 2933	31–1989 (CLIP) 45–1989	U213, R211, R212, C146, C147, L223 added CLIP ERC 2928: R184 619k (was 1M), R185 3.01M (was 4.7M) R112 3.01k to 5.62k, U112 LM6364N to AD848JN CLIP ERC 2931: R142 & R143 added Later (45–1989): JM183 replaced with R182 140Ω || R183 10kΩ
66503	C	2949 3011 3047	50–1989 10–1990 (highest DC) 22–1991	TBD Later (10–1990): R172 402Ω (was 3.48kΩ) Later (22–1991): R182 100Ω (was 140Ω), R183 1k (was 10k), R141 18.7k (was 12.1k), R142 1k (was 10Ω)
66503	D	3519	19–1995 (highest DC)	R144 16.7k added R143 51.1k (was 1k) & connection to −15V (was XCOMP) CR144 & CR145 added U213 removed (short between pin 8 & 10) U230 05C metal case (was plastic) U404 replaced with LT1122 (was LF400C) Q401, Q402 replaced with TO–92 2N3906 (was TO–18 2N4917)
66513	A	4243 ? ?	41–2000 (highest DC) 47–2000 (highest DC) 36–2004 (highest DC)	U210 Gate array replaced with Orbit 6968A (was Fujitsu MB651314) U301 & U351 replaced with HFBR–2521Z (was HFBR2501) U304 & U354 replaced with HFBR–1521Z (was HFBR1510) R122 301Ω (was 243Ω, latest 47–2000 – TBD change on which REV) Later (36–2004): U142/U405 patchboard 66530: EL2252CM (was 2xEL2018CN)
66513	B	?	22–2016 (highest DC)	U210 Gate array patchboard 26550: Altera MAX V 5M1270ZT144C5N (was Orbit 6968A) + level shifter SN74LVC8T245P + LDO LMS8117AMP–1.8 & LMS8117AMP–3.3 U220 µC replaced with ATMEL AT89C51 (was Intel P80C51BH) U111 & U140 replaced with AD817AN (was LM6361N)
66523	2	? wekomm	29–2019 42–2019	Full redesign in RoHS SMD U142, U181, U405 replaced with TL3016C (was EL2018CN/EL2252CM) + discrete level shift & ±5V from 78/7905 regulator U220 µC replaced with ATMEL AT89LS51 Low–Voltage (was AT89C51)
66533	002	? Keysight	04–2021 30–2021	C119 150p || 180p (was 1x330p) & C409 470p || 10p (Σ480p, was 1x510p) R407 47Ω (was 100Ω) JM210 changed to SMD 0Ω (was wire bridge) Time interpolator: added some 0Ω in series
66533	003	? Keysight	38–2025	TBD

JACK TO DC BOARD

ADC REFERENCES

IC–A/D HYBRID

HYBRID

Consists of two dies:
Resistor network (23 resistors)
Digital IC:
– decoder logic
– 18 driver & dual MF–MOSFETs (5x dummy)
– FET modulation (single MF–MOSFET)

Supply for switch drivers rusn1&2 (instead of vdd2)

Supply for switch drivers (except rusn1&2)

Supply for logic (except switch drivers)

Mod. switch sense = source? gnd = drain? vdd2 = gate?

INPUT FET MODULATION

INTEGRATOR

SLOPE AMP

COMPARATOR

MINOR SLOPE DRIVERS

ZERO GLITCH JUMP CIRCUIT

Deviation to original: XS64 to Pin 3, S16 to Pin 5 of U150
(Routing to U180 suboptimal, easier to just relocate vias)

AD848 compensation

Buffer for U111/U112

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